Choosing and Building Clusterings: K-Means Model Selection and Agglomerative Clustering from Scratch

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Abstract

We study two families of clustering algorithms. For K-means, applied to 2,360 two-dimensional points, we choose the number of clusters with the elbow method and cross-check it with the silhouette score. Both point to $k = 5$ (WCSS 611.9, silhouette 0.580). However, the WCSS curve has a second knee at $k = 7$, and the $k = 7$ solution recovers seven visibly separate groups that $k = 5$ merges in pairs. Neither criterion alone is decisive here.

For hierarchical clustering, we implement bottom-up agglomerative clustering from scratch with single, average and complete linkage, and verify that all 19 merge heights match SciPy exactly. On 20 one-dimensional points, average and complete linkage agree on the same two- and three-group partitions. Single linkage shows its well-known chaining *behaviour: its final merge happens at a distance of only 10.7, against 94.9 for complete linkage.*

1. Introduction

Clustering has no labels to check against, so the main difficulty is not running an algorithm but deciding which result to trust. We look at two such decisions: *how many* clusters to use, and *how to measure distance between groups*.

2. Related Work

K-means [3] minimises the within-cluster sum of squares (WCSS), and k-means++ [1] seeds it with well-spread centres. The elbow method looks for the k beyond which WCSS stops falling sharply. The silhouette score [4] compares each point's distance to its own cluster with its distance to the nearest other cluster. Agglomerative clustering [2] repeatedly merges the closest pair of clusters. Single linkage uses the closest pair of members, which makes it prone to chaining; complete linkage uses the farthest pair, favouring compact clusters; average linkage sits between them.

3. Method

3.1. K-means

The two features are standardised, and K-means (k-means++ seeding, 10 restarts) is run for $k = 1, \ldots, 15$. For each k we record the WCSS, its relative drop from $k - 1$, and the mean silhouette. Cluster centres are mapped back to the original units for plotting.

Table 1. Model-selection statistics around the elbow.

k	WCSS	Drop vs. $k-1$	Silhouette
3	1506.6	45.3%	0.495
4	858.6	43.0%	0.571
5	611.9	28.7%	**0.580**
6	459.0	25.0%	0.562
7	365.0	20.5%	0.561
8	332.4	8.9%	0.539

3.2. Agglomerative clustering

Every point starts as its own cluster. At each step we compute the linkage distance between all pairs of current clusters, using Euclidean point distances δ:

$$D_{\min}(A, B) = \min_{u \in A, v \in B} \delta(u, v),$$
$$D_{\text{avg}}(A, B) = \text{mean}_{u \in A, v \in B}\, \delta(u, v),$$
$$D_{\max}(A, B) = \max_{u \in A, v \in B} \delta(u, v).$$

We then merge the closest pair and record the merge in a SciPy-format linkage matrix so that dendrograms can be drawn. Correctness is checked by comparing all merge heights against `scipy.cluster.hierarchy.linkage`.

4. Results and Discussion

4.1. How many clusters?

The elbow and the silhouette agree on $k = 5$. WCSS falls by over 40% at each step up to $k = 4$, and by less from $k = 5$ (Table 1). The silhouette peaks at $k = 5$. The $k = 5$ solution assigns 490, 464, 740, 274 and 392 points to its five clusters (Fig. 1).

But the data suggest seven groups. The relative WCSS drop stays at 20–25% through $k = 7$ and only then falls below 10%, which is a sharper knee than the one at $k = 5$. In the $k = 5$ solution the largest cluster (740 points) and another (490 points) each visibly contain two blobs. The

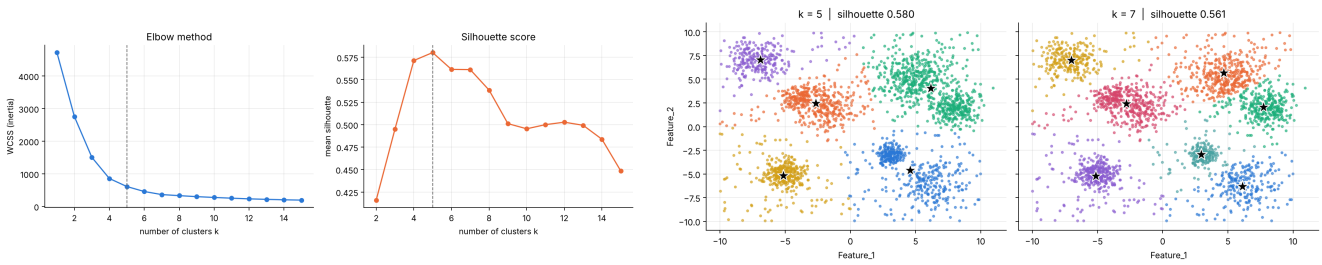


Figure 1. Left: WCSS (elbow) and silhouette score for $k = 1$–15; dashed lines mark $k = 5$. Right: K-means solutions for $k = 5$ and $k = 7$ (stars are centres). With $k = 7$ each visible group gets its own cluster.

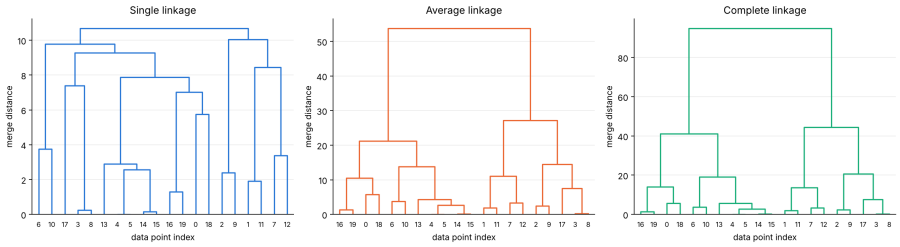


Figure 2. Dendrograms from the from-scratch implementation. Note the very different vertical scales: single linkage merges everything below 11, complete linkage only at 95.

$k = 7$ solution separates them (Fig. 1, right). Its silhouette is only slightly lower (0.561 vs. 0.580) because the merged pairs are adjacent, so treating each pair as one cluster costs little separation.

So $k = 5$ gives the best-separated coarse grouping and $k = 7$ the finer structure; without ground truth, both should be reported.

4.2. Hierarchical clustering: linkage matters

Correctness. For all three linkages, the 19 merge heights produced by our implementation match SciPy exactly.

Average and complete linkage agree. Cutting either dendrogram into two groups gives $\{2.1 \ldots 43.2\}$ (11 points) and $\{52.5 \ldots 97.0\}$ (9 points). Cutting into three splits the upper group at the gap between 73.2 and 83.2. The large final merge heights (53.8 and 94.9) show well-separated top-level groups (Fig. 2).

Single linkage chains. Because it joins clusters through their closest members, single linkage links the 1-D points one gap at a time, and every merge happens between 0.002 and 10.7. Its two-group cut is lopsided: 14 points up to 60.1, then the 6 largest values, split at the widest single gap (60.1 to 70.8). This is the textbook chaining effect: single linkage finds the largest gaps but ignores cluster compactness.

5. Conclusion & Future Work

The elbow and silhouette criteria both favour $k = 5$, but a second elbow and the scatter itself point to seven groups, so model selection should be reported with its ambiguity rather than as a single number. The from-scratch agglomerative algorithm reproduces SciPy exactly and shows the expected contrast between chaining single linkage and compact complete linkage.

Next steps. (i) Add the gap statistic [5] and Gaussian-mixture BIC as further criteria for k; (ii) assess the stability of the $k = 5$ and $k = 7$ solutions under bootstrap resampling; (iii) speed up the agglomerative implementation with a priority queue or the Lance–Williams update; and (iv) apply it to the 2-D dataset to compare against K-means directly.

Acknowledgement. This work was carried out as part of EE708 Fundamentals of Data Science and Machine Intelligence at IIT Kanpur under the supervision of Dr. Rajesh M. Hegde.

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